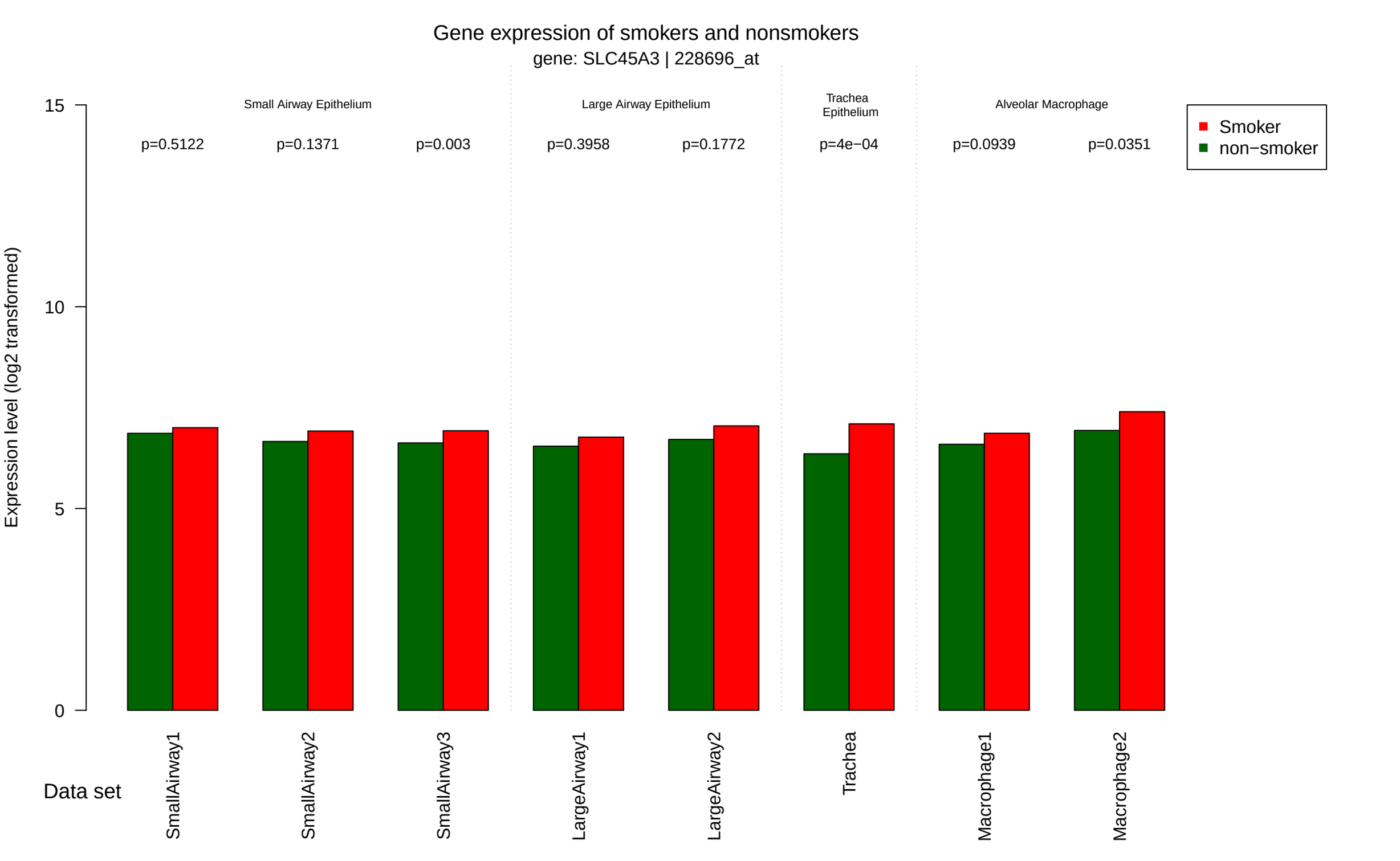
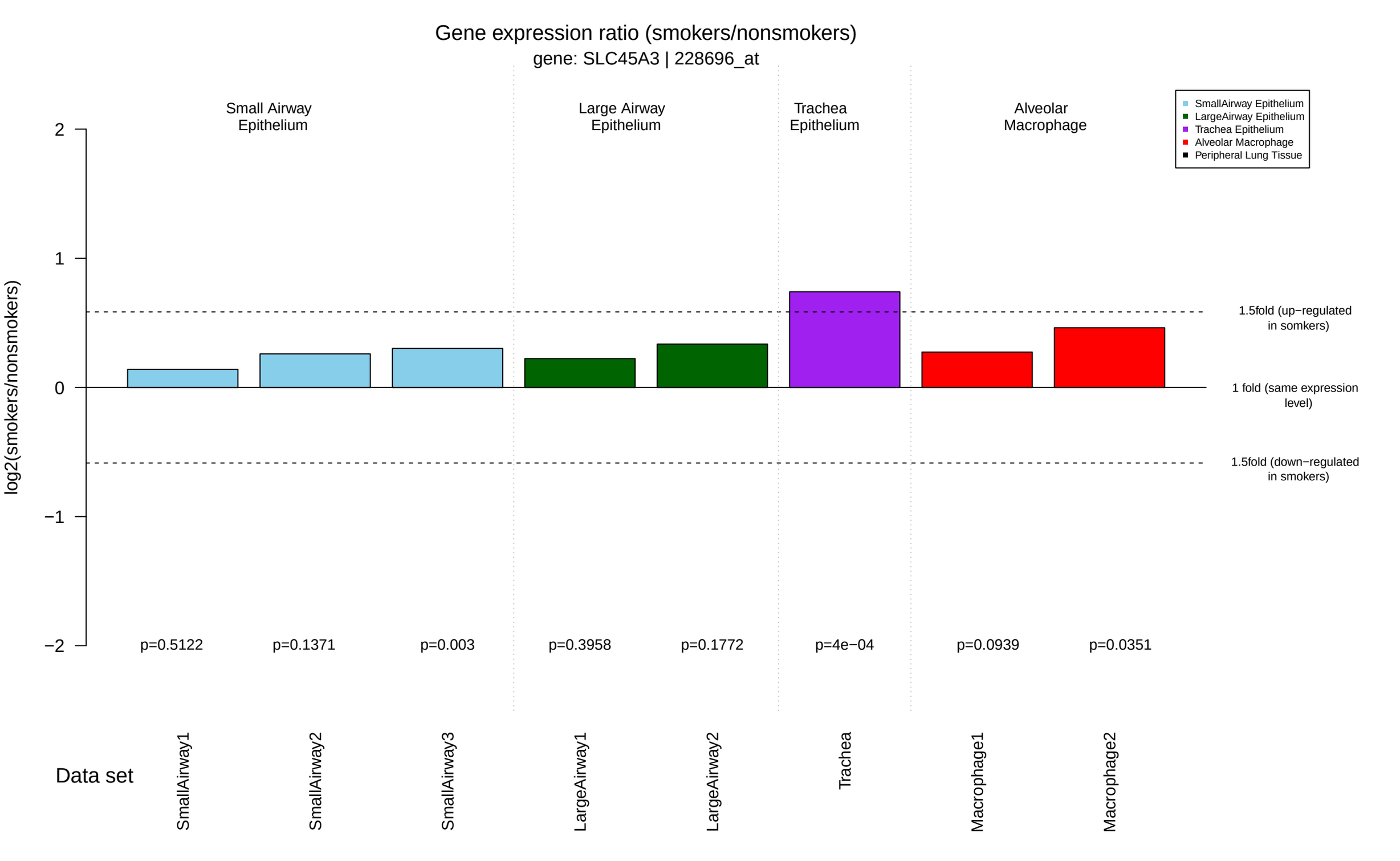
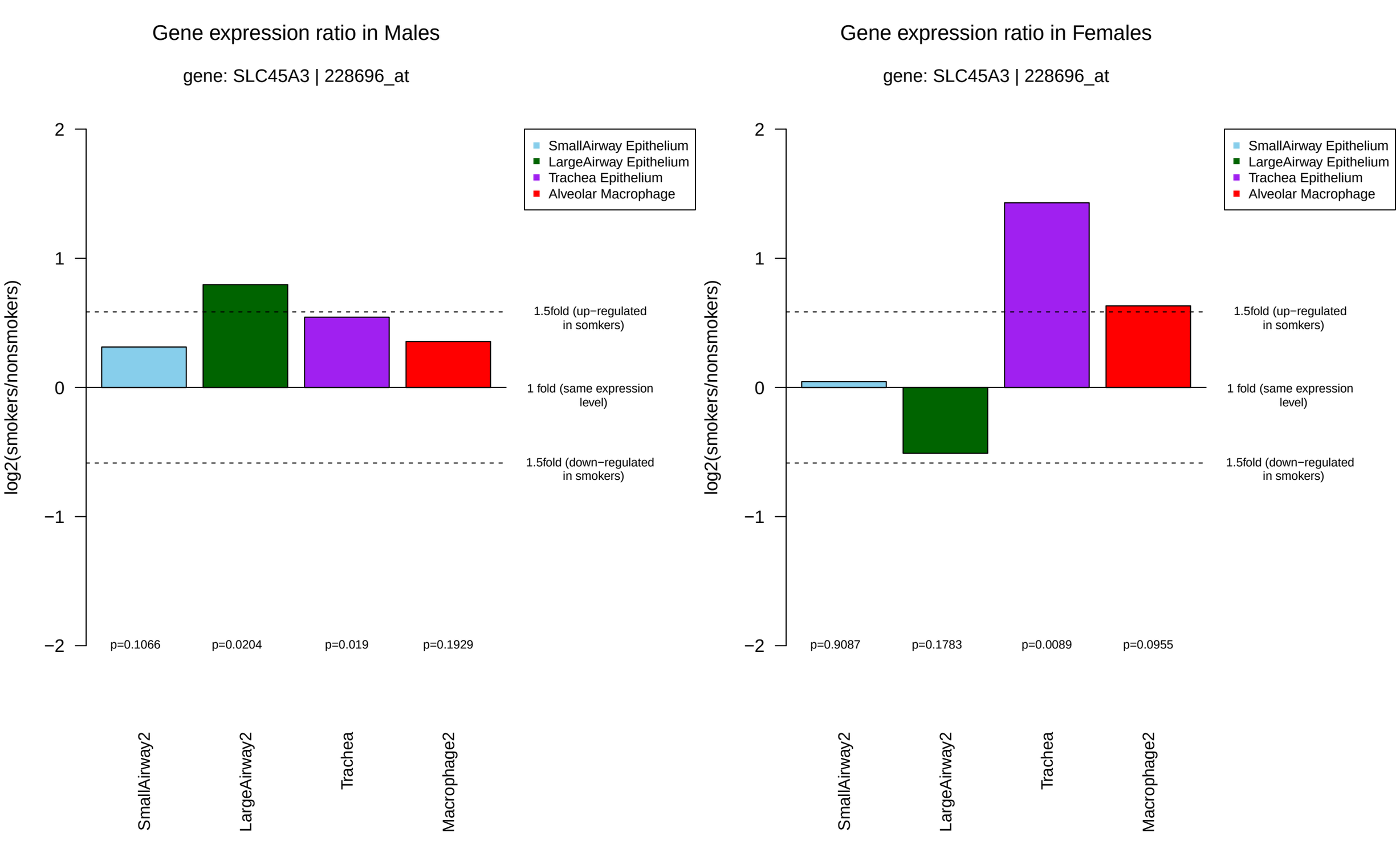
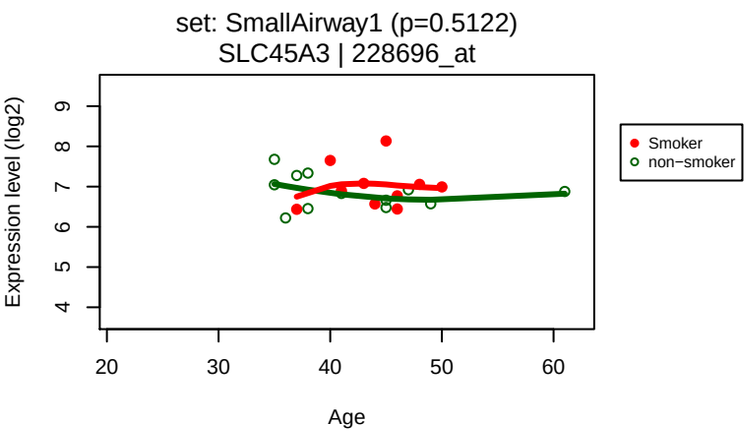
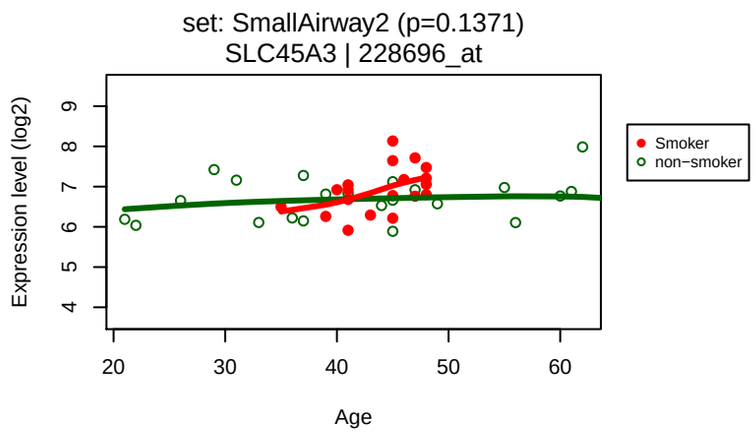
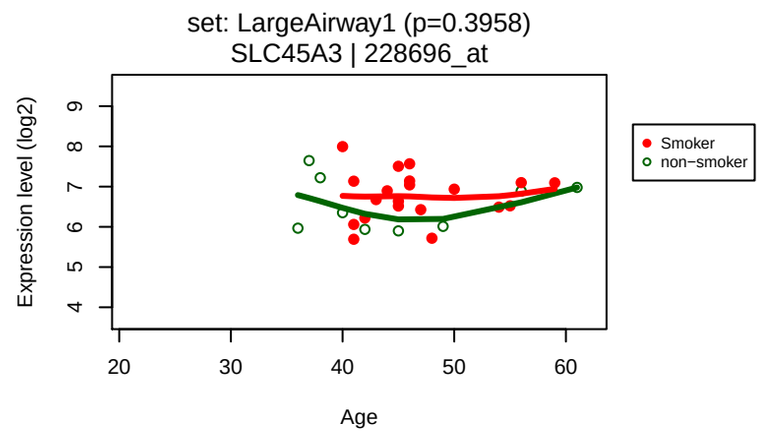

Gene expression of smokers and nonsmokers
gene: SLC45A3 | 228696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5122
p=0.1371
p=0.003
p=0.3958
p=0.1772
p=4e−04
p=0.0939
p=0.0351
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

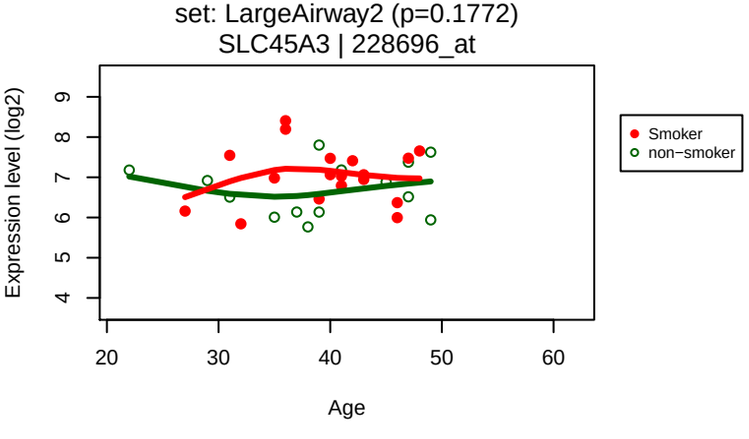

Gene expression ratio (smokers/nonsmokers)
gene: SLC45A3 | 228696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5122
p=0.1371
p=0.003
p=0.3958
p=0.1772
p=4e−04
p=0.0939
p=0.0351
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

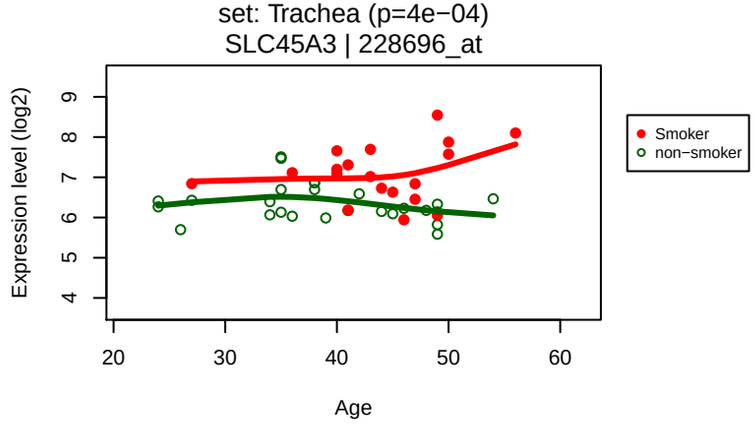

Gene expression ratio in Males
gene: SLC45A3 | 228696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1066
p=0.0204
p=0.019
p=0.1929
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC45A3 | 228696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9087
p=0.1783
p=0.0089
p=0.0955
SmallAirway2
LargeAirway2
Trachea
Macrophage2

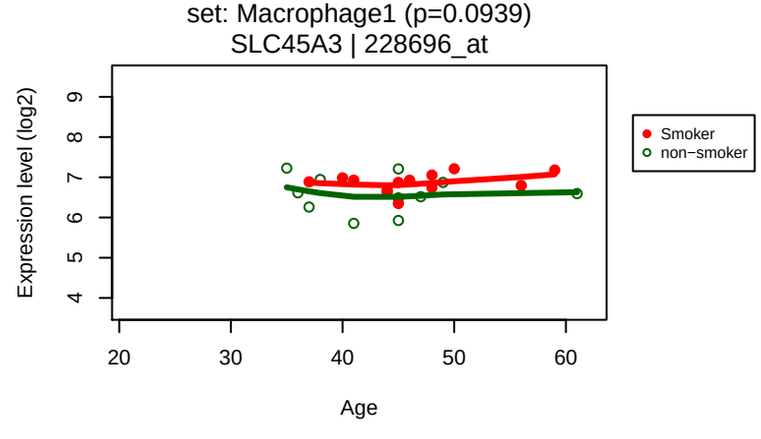

set: SmallAirway1 (p=0.5122)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker

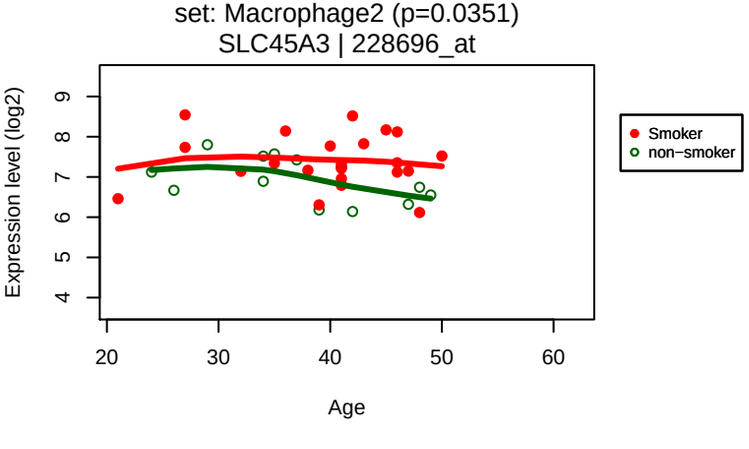

set: SmallAirway2 (p=0.1371)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker

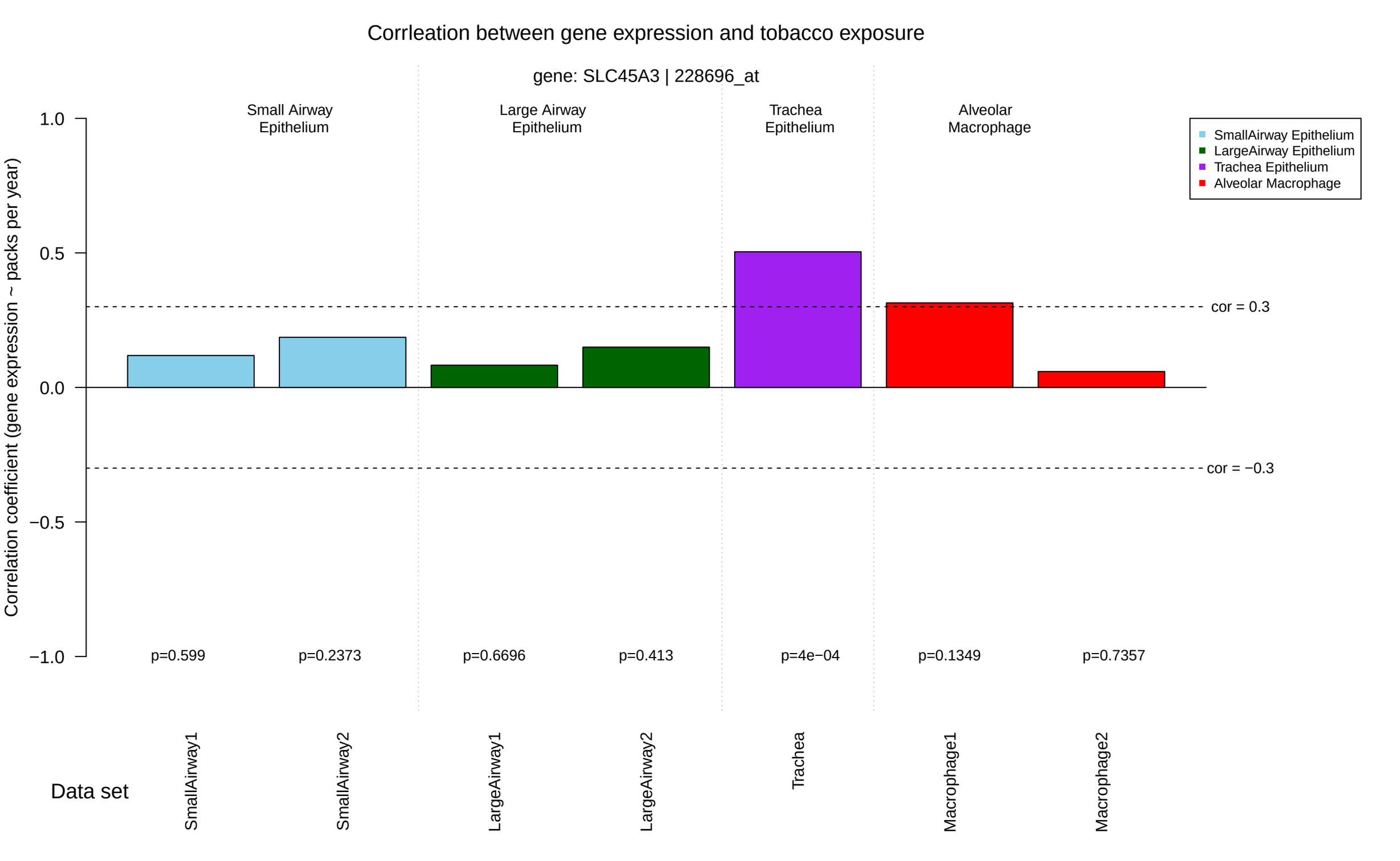

set: LargeAirway1 (p=0.3958)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1772)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0939)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0351)
SLC45A3 | 228696_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC45A3 | 228696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.599
p=0.2373
p=0.6696
p=0.413
p=4e−04
p=0.1349
p=0.7357
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2